

PACKED WORD LAYOUT

word = (counter0..1023 ×32) + (code0..6 ×2) + permit0/1

Bit0: permit; bits1..4: judgment code; bits5..14: counter; bit15 must be0. Invalid sentinel32767 is rejected.

LS ENVIRONMENT RECEIVER DEVICE MAP	
LS device	Function
D00400	CIMON destination write
D00401	PLC-owned RX echo read-back
D00402	Permit bit
D00404	Judgment code
D00407	Decoded live counter
M00400	Frame format valid
M00405	Valid + received change + age fresh
M0100C	Final environment start permit

BRIDGE AND RECEIVER GUARDS

CIMON checks the decoded D100 live counter for freshness.
LS D00030 live counter supervises LS progress.
Bridge confirms D00401, not its own write-cache value.
LS T0010 preset50 (100ms):5s unchanged receiver closes permit.
Current application bridge FX no-change threshold3s.

Counter change proves publisher progress, not a new sensor sample. FX separately checks BOOT/SEQ/status/AGE and recovery samples.

Bridge failure clears data-valid displays and blocks future cycles. Restoring communication requires controlled bridge restart and a new process START.

NO.	DATE	REVISION
R0	2026-10-22	FUNCTION RECORD

DWG TITLE

SEMICON SORTER PROJECT
Final Test and Multi-Bin Sorting Cell

FX TO LS ENVIRONMENT TELEGRAM

DRAWN BY	DATE
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DWG NO
013

SHEET NO
1 OF 1